Ethereum Fraud Detection

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PART 1

INTRODUCTION

TOPIC INTRODUCTION

Fraud Features

Identify critical fraud features

Model Building

Build high-accuracy classification model

Algorithm Comparison

Compare algorithm performance, select optimal solutions

PROJECT OVERVIEW

Background

Background

- Blockchain popularity → Increase in Ethereum fraud → Users asset loss
- Inefficient manual detection → Need for automated, intelligent solutions

Core Content

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- EDA-based feature selection and data preprocessing
- Handle outliers, missing values, and data imbalance issues
- Train and compare 6 models including LR, XGBoost, SVM and etc.

Significance

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- Technical support: Ensure blockchain security, reduce fraud risk
- Method exploration: Provide a reference for feature engineering and modeling in high-dimensional data scenarios

DATASET OVERVIEW

Basic Information:

- **Dataset Name:** **Ethereum Fraud Detection Dataset**
- **Source:** **https://www.kaggle.com/datasets/vagifa/ethereum-frauddetection-dataset (Kaggle)**

Scale & Structure:

Raw Data
9,841 records
48 features

Label
FLAG :
0 = Legitimate, 1 = Fraudulent

Distribution
Imbalanced
7,662 vs. 2,179(fraud)

DATASET OVERVIEW

Key Features:

Time

- Avg. interval
- First-last time diff

Volume

- Sent / Received txns
- Contracts
- Unique addresses

Aoumnt

- Min / Max / Avg ETH sent / received
- Total amount

ERC20 Tokens

- Txns
- Token amounts
- Unique token names

MAIN CONTENT

Data Quality Note:

Missing Value

829 records（~8.4% missing rate）

Some have null values, requiring filling or deletion

Duplicate Record

546 rows

Caused by entry errors or system resubmissions

Extreme Outliers

High amounts and long intervals

PART 2

DATA PROCESSING, PCA

DATA PROCESSING, PCA

Phase 1: Initial Cleaning & Handle Missing Values

- **Remove duplicates and low-variance（std < 0.001） features**
- **Missing Values**

Feature Type	Handling Method	Strategy Explanation
Numerical features	Impute with mean	Preserve data distribution
High-cardinality categorical features	Impute with mode	Missing rate up to 73%

Phase 2: Detect and Treat Outliers

- **Business Rule-Based Outlier Removal**

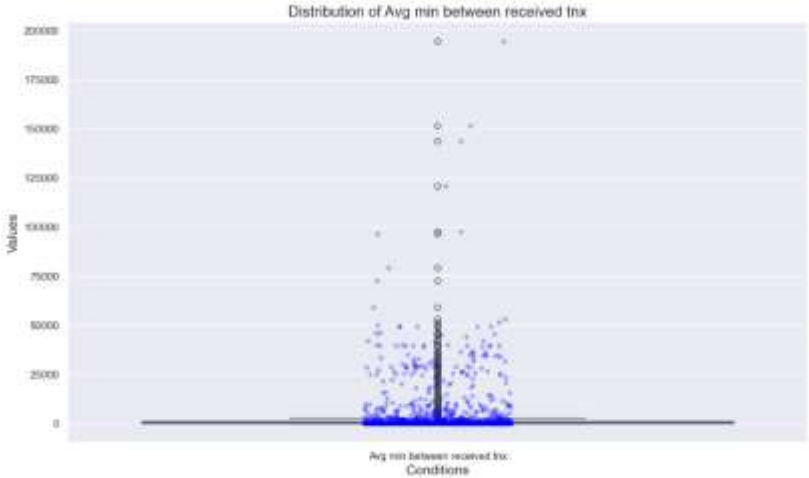
Remove outliers from fraudulent transaction samples (FLAG=1) :

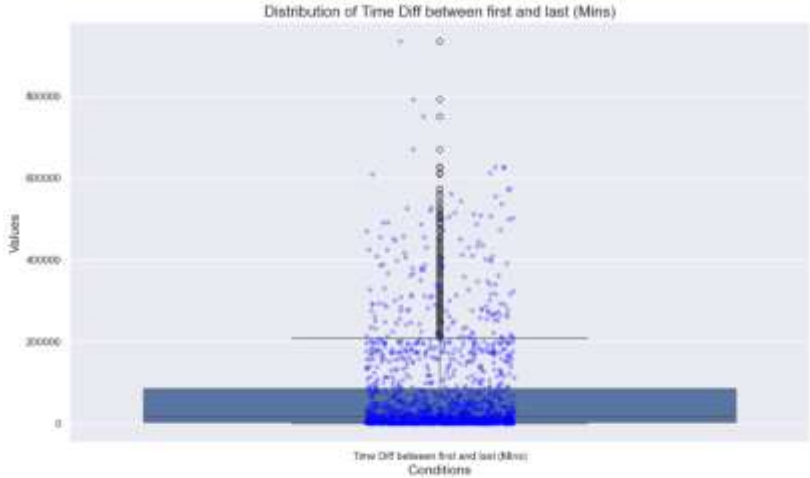
Feature	Threshold
Time Diff between first and last (Mins)	> 600,000
Avg min between received tnx	> 60,000
Received Tnx	> 8,000

DATA PROCESSING, PCA

- ## Visualization

Scatter plots reveal samples far from the main distribution area:





- ## IQR-Based Outlier Treatment

Calculate IQR = Q3 - Q1 for each feature

Outlier range: outside [Q1-1.5 × IQR, Q3 + 1.5 × IQR]

Outlier ratio ≤ 7% ➡ Remove

Feature	Ratio	Result
avg val sent	6.59%	Removed
ERC20 total Ether sent contract	0.32%	Removed
ERC20 uniq sent addr.1	0.00%	Removed

DATA PROCESSING, PCA

Phase 3: Feature Engineering & Data

- ## Imbalanced Data

Fraud transactions are the minority class.

Evaluation focuses on:

Precision – Accuracy of fraud predictions
Recall – Coverage of actual fraud cases
F1-Score – Harmonic mean of both

- ## High-cardinality categorical features:

ERC20_most_rec_token_type: 467 unique values
ERC20_most_sent_token_type: 305 unique values

Label Encoding:

Map each category to a unique integer,
converting text features into numerical format suitable for modeling.

DATA PROCESSING, PCA

- **Feature Scaling & PCA**

PCA Dimensionality Reduction:

- Scale features to zero mean and unit variance
- Retain 95% of variance information

Metric	Value
Original Feature Dimension	37
After PCA Compression	21 principal components
Further Selection	8 principal components

Isolation Forest Outlier Detection:

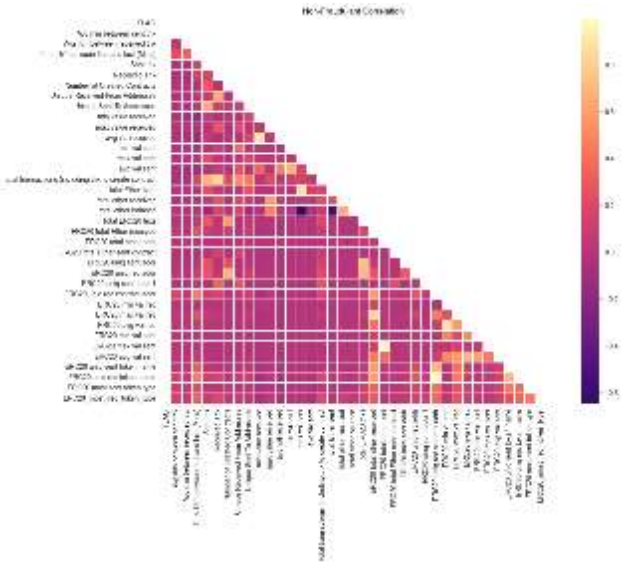
- Isolation Forest algorithm
- Assume 5% outliers for further anomaly removal

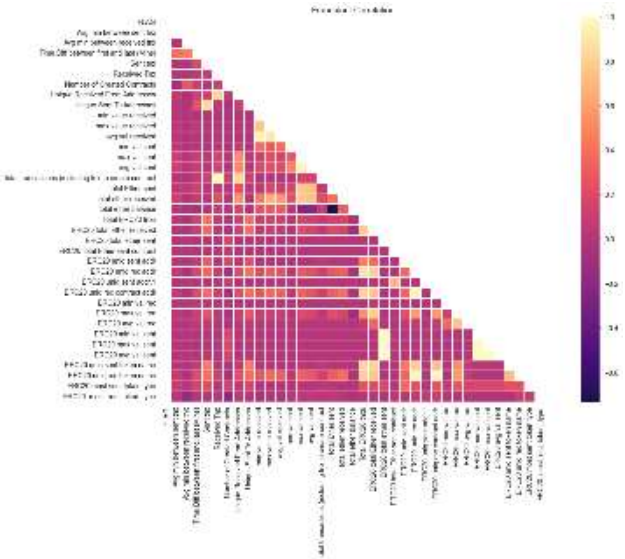
Item	Before	After	Change
Sample Count	9,295	8,830	465 rows removed (5%)

SUMMARY

Correlation Analysis

- Correlation heatmaps separately for fraud and non-fraud transactions
- Upper triangle masking to avoid duplicate display





Limitations & Future Improvements

- PCA strengths:

Data Characteristics	Dimensionality Reduction	Model Compatibility

- Improvement Direction:

  Loading matrix for interpretability, Dual-path comparison

PART 3

MODELS SETUP

MODELING PROCESS

1. **Data split: 70% training / 30% testing**

2. **Feature scaling → MinMaxScaler**
   Dimensionality reduction → PCA (95% variance retained)

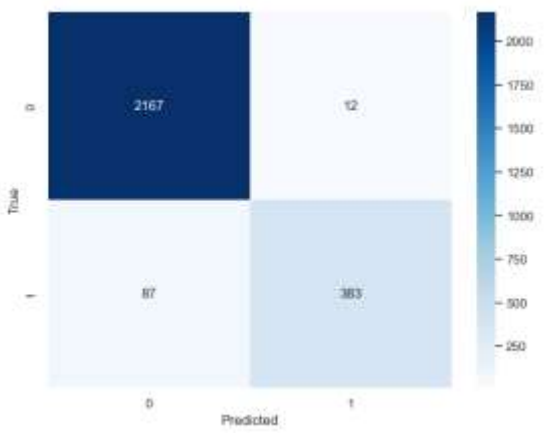
3. **Evaluation metrics: precision, recall, F1-score (due to class imbalance)**

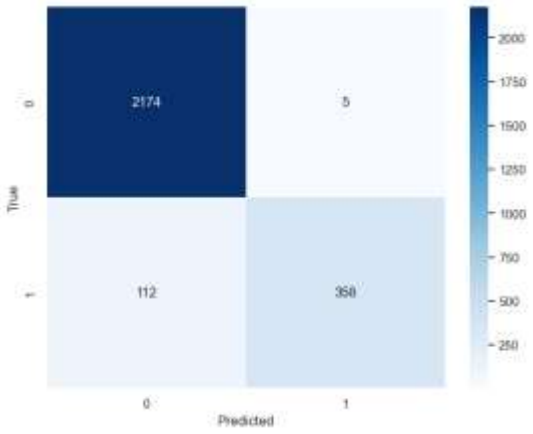
COMPARATION

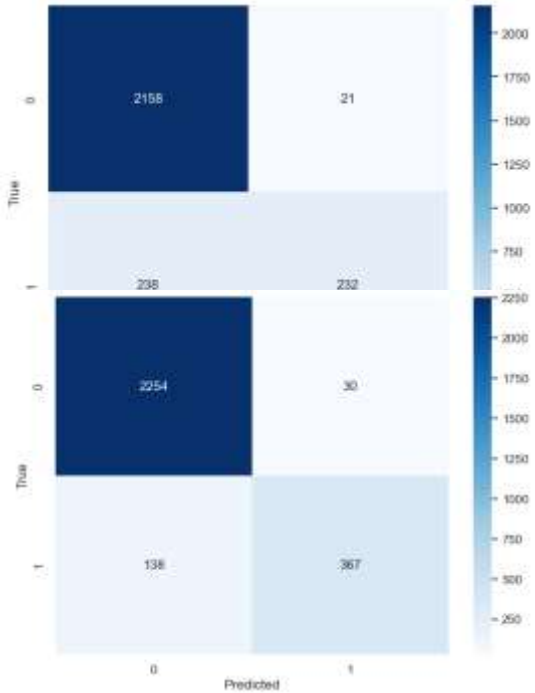
Model	Precision (Fraud)	Recall (Fraud)	F1-Score (Fraud)	Accuracy
Logistic Regression	0.92	0.49	0.64	0.90
XGBoost	0.97	0.81	0.89	0.96
Random Forest	0.99	0.76	0.86	0.96
SVM	0.20	0.97	0.33	0.31
AdaBoost	0.92	0.73	0.81	0.94
Naïve Bayes	0.20	0.97	0.33	0.31

XGBoost is recommended as the final model for this fraud detection task due to its superior F1-score and balanced performance across both classes.

Confusion Matrix Highlights







XGBoost

– True positives (fraud correctly identified): high
– False positives (normal misclassified as fraud): low
– Best overall balance

Random Forest

– Very high true positives
– Extremely low false positives → conservative

SVM & Naïve Bayes

– Very high false positives → many normal transactions flagged as fraud
– Impractical for real use

Logistic Regression & AdaBoost

– Moderate false positives
– Lower true positives than XGBoost

ANALYSIS

- **XGBoost achieved the best balance between precision and recall, making it the most reliable model for fraud detection (F1 = 0.89).**
- **Random Forest had the highest precision but lower recall, meaning it was cautious about labeling transactions as fraudulent.**
- **Logistic Regression performed moderately well and served as a solid baseline.**
- **SVM and Naïve Bayes had very low precision for the fraud class, indicating they misclassified many legitimate transactions as fraud.**
- **AdaBoost performed well but slightly worse than XGBoost.**

PART 4

CONCLUSION

PROJECT SUMMARY

We built a fraud detection system for Ethereum transactions using a comprehensive pipeline:

Data preprocessing

Missing values, duplicates, outliers, categorical encoding

Feature engineering

Removed low-variance & highly correlated features.

Dimensionality reduction

PCA (95% variance)

Modeling

Six classifying models evaluated (Focus : precision / recall / F1 score)

KEY FINDINGS

- **XGBoost best model (F1 = 0.89)**
- **Preprocessing steps (outlier removal, feature selection) critical**
- **Machine learning effective for Ethereum fraud detection**

FUTURE IMPROVEMENTS

Additional features – transaction timing, user behavior, network volatility

Class imbalance – SMOTE or other sampling techniques

Hyperparameter tuning – further optimize XGBoost

Deployment – real-time monitoring with live blockchain data

2026

THANK YOU